

Water Level and Water Quality Monitoring System for Water Tanks

Project proposal – CTP 2021

Project ID: CTP/2021/07

Group members:

- | | |
|--------------------------|----------|
| 1. Perera H.A.K.D | 20543912 |
| 2. Kasthuriarachchi S.D. | 20543873 |

Supervised by:

.....

Dr. Anuradha Jayakody

.....

Date of Submission

Abstract

One of the major issues in the industry and common households is the proper management of water tanks. Water is a key part of our day to day lives and proper management and storage of water is necessary. The system proposed in this document provides a solution to proper management of water levels and testing of the quality of water stored in tanks. Most households and small-scale industries in countries like Sri Lanka and India use overhead tanks for storing water and Improper management of these water tanks lead to wastage of large amounts of water and technical difficulties such as dry running of pumps and excess tank pressures. This system allows real time monitoring of water quality and the water levels, thereby turning pumps on or off depending on the current water level of the tank. The users logged into the system will receive alerts depending on the water level While the water level could be checked using NODMC, GPRS also could be used in case of a failure in the NODMC, making it more productive to prevent water wastage. The water level in the tank could be measured using an ultrasonic sensor which is connected to a WIFI chip in the water tank, through the WIFI network. In the event of WIFI-network failure, the defaults could be obtained through an auto GPRS and vice versa. This device can fill water up to maximum capacity of the tank without wastage and automatically sends alerting messages and can turn ON motors when the water limit is low and switch OFF motor when it is high so that water overflowing could be prevented. This component can also be used to measure levels of any liquid stored in tanks other than water as well but the water quality monitor only functions to test quality of water. The water quality component is able to get readings of main qualities of water such as turbidity, pH levels and dissolved oxygen present in the water. Therefore, the main objective is to provide homes and small-scale industries a system to monitor water levels in tanks and minimize water wastages technical difficulties related to water containment.

Table of Contents

Abstract	i
Table of Content.....	ii
List of Figures	iii
List of Tables	iv
Chapter 1: Introduction	1
1.1 Introduction to the project	1
1.2 Problem identification	2
1.3 Objectives	2
1.4 Research Questions	3
1.5 Significance of the study.....	3
1.6 Aims of the project.....	4
Chapter 2: Literature review	5
Chapter 3: Methodology.....	8
3.1 Planning	8
3.2 Technologies to be used.....	8
3.3 User Requirements	13
Chapter 4: Testing and Evaluation.....	18
4.1 Testing.....	18
4.2 Evaluation	18
Chapter 5: Personnel and facilities.....	19
Chapter 6: Budget.....	20
Chapter 7: References.....	21

List of Figures

Figure 2.1: Wireless Tank Level Monitoring	5
Figure 2.2: Mobile Tanker Level Monitoring	6
Figure 2.3: Water quality System Accessibility diagram	7
Figure 3.1: Circuit Diagram of Water Quality Measuring Component	9
Figure 3.2: Circuit Diagram of Water Quality Measuring Component	9
Figure 3.3: NodeMCU	10
Figure 3.4: Sim 800 GSM IC CHIP, Sim800	11
Figure 3.5: Ultrasonic Sensor	12
Figure 3.6: Data Transfer Cable	12
Figure 3.1 Accessibility Diagram	14
Figure 3.2: Flow chart water quality measuring component	15
Figure 3.3: Software that may be used	17

List of Tables

Table 3.1: Timeline.....	15
Table 10.1: Budget.....	18

Chapter 1: Introduction

1.1 Introduction to the project

This project proposal includes the information regarding a system to manage water quality and water levels of tanks. This system is able to monitor the status of water levels in water tanks at real time and also check the quality of water and display them in a centralized web application. This system has a number of features such as automatic alerting of users regarding water levels, controlling or displaying status of the pumps. This system will be built as two separate components, one component for measuring water levels and another to check for water quality. Further details regarding the features, system specifications and objectives will be discussed in this document.

1.2 Problem identification

In countries like Sri Lanka and India and most similar countries wastage of water has become a major issue and even the quality of water stored in tanks may not be up to desired level of quality. In most cases people are unaware of the water levels of tanks and fill tanks until they overflow causing major water wastages. This happens not only in domestic households but also small-scale industries as well. Also, in most cases people pump water from wells (underground water) to overhead tanks to be used when necessary, mostly because certain areas of the country doesn't have water supplies from government at the moment. Through this proposed system we are able to monitor the real time water levels of water tanks at any given time and send alerting messages to the users depending on water levels of tank. This eliminates the risk of dry running of pumps due to low water levels present in tanks, high water pressure differences due to excess pumping of water and also helps minimize wastage of water because the pumps can be switched on or off automatically. This system also provides a solution checking the quality of water thereby ensuring that clean water is present in the tank at all times. This system will check the qualities of water such as turbidity, pH and dissolved oxygen levels and display them in the web application.

Main features of the system

- Water quality and current water levels can be monitored in real time from one centralized system.
- Water can be filled to any desired level.
- The system can be controlled from any location once the user has accessed the system
- Alerts users to switch off pumps when maximum water level is reached and switch them back on when the water levels have reached the minimum required levels.
- Pumps can be automatically switched on or off depending on the water level in the tanks

Advantages of the system

- This system helps minimize water wastage significantly due to automatic control of water pumps
- Can monitor water levels and fill tanks to any desired level.
- The quality of water stored in tanks can be measured easily.
- The users have access to control the system from any location.

This system consists of mainly two components. One component will be used to measure water level in the tanks while the other component is used to measure water quality. There are several reasons for separating the device into two components and the main reasons are listed below.

- The water level measuring component must be fixed at the top of the water tank
- The water level measuring component can also be used to measure levels of other liquids stored in tanks as well.
- The water quality measuring component can be fixed in any location of the tank if required.

This system prevents overflowing of water by giving an automatic alert and indicating that the water has reached the maximum level and automatically switches off the pumps when maximum capacity is reached. A Wi-Fi network called NODEMC will be created which could be accessed through an alert. An ultrasonic sensor will be connected to a WIFI chip in the water tank through which, the water tank level could be measured using the sensor by using the Wi-Fi network. In case the WIFI-network failure, the defaults could be obtained through an auto GPRS and vice versa. The motor is also able to automatically turn ON when the water limit is low which helps prevent dry running of pumps in most industries and can be a helpful feature in domestic households as well.

1.3 Objectives

Most people are used to switching off water pumps when they observe that the tanks are overflowing. This causes major wastages of water and other resources such as electricity. With the busy lifestyle of people, they do not have time to wait around a switch to start and stop water pumps whenever they must fill water tanks. This proposed system gives the freedom to do other work and save their time because this system can be accessed by users from any location and can monitor status of water levels and pumps and also start and stop pumping of water when required. If a user forgets to turn off a pump the system automatically alerts the user when the maximum water level in the tank is reached. Similarly, an alert is sent if the water level is below required levels as well.

The main objectives of the system are mentioned below,

System objectives

- Analyse people's needs regarding containment of water in countries like Sri Lanka.
- Improve and maintain the system to display and manage water tanks.
- Analyse user requirements when storing water.
- Analyse suitable solution and development technologies to overcome problems regarding storage of water.

Outcome objectives

- Prevent wastage of water by at least 60%
- Provide a centralized system for users to view status of water levels, quality of water and pumping status of water to tanks
- Make pumping of water to tanks and view status of tanks easier.

1.4 Research Questions

In most countries like Sri Lanka and India most people do not have access to the public water supply provided by the government. Most people pump water to overhead tanks to be used when necessary. Pumping water to overhead tanks requires water pumps that can pump water to the desired height and proper tanks to store the water until required. Generally, people use well water (underground water) or water from streams or other water bodies and these water bodies may contain impurities or chemicals as well. So, some of the questions under this study are how many people in Sri Lanka do not have access to public water supply and how much of them use tanks for water storage. The status of water, whether is in good quality or not is yet another major factor to be considered. Finally, questioning of people whether they would benefit from this proposed system or what sort of additional features they may require were one of the questions that came up under this study.

1.5 Significance of the study

It has been a major issue nowadays that huge amounts of water have been wasted due to water overflowing in tanks at houses and small scale industries. This project intends to reduce the water wastage by 60% upon successful implementation. In most cases, people are unaware till the water tank has overflowed or what quality the water is in, but through this system users can accurately monitor the water level in the tank from any location, in real time through a web application and manage the water level and quality, so that water will not be wasted by overflowing. For testing of water level, most often contactless strategies are optical technique, radar, and ultrasonic technique. Since we do not want to affect the top-notch of water within the tank, we put into effect one of the contactless strategies. Most contactless strategies work on the same precept; Sending a signal and wait for a degree the time that the sent signal returns to the device. The optical techniques use optical signals and may be the most accurate method to test the water level; However, sensors can get dirty over time and prevent total functionality of the device. The

ultrasonic approach is similar to radar technology however, instead of radar waves, we are sending ultrasonic waves. This project will be using ultrasonic approach and this device is right for our purposes since ultrasonic sensors are less costly. The water quality measuring component will be measuring the main required qualities of water. Since the quality of water is a broad topic and consists with many factors to be considered, this project will only be considering the most important factors such as Turbidity, dissolved oxygen levels and pH levels. The significance of this project is that it gives the users of the system, a centralized platform (Web Application) to monitor water levels and water quality of the water stored in tanks at real time. This system is also able to minimize water wastage significantly by sending alerts to users when water level has reached maximum capacity and similarly send alerts if water level is too low as well. The users of the system can also easily view the status of the pumps, whether it is in pumping or stopped status and users have the freedom to start and stop pumps through the system easily when required.

1.6 Aims of the project

This project has a number of aims and goals and they are listed below,

- provide a system that every house and company can use to manage water tank for their storage purposes.
- Alert and notify users when the water level has exceeded maximum levels or when it is below minimum levels to start and stop pumps as required.
- Provide users the ability to fill up the tank to their own preferred level.
- Develop a platform (Web Application) where users can monitor status of water pumps, water levels or water quality at real time.
- When the water limit is high, the user gets an alert.
- The pump status could be observed or turned ON and OFF as needed from any place via the Web Application.
- This system's main objective is to prevent water wastage and maintain quality of water in tanks which store water.

Chapter 2: Literature review

- Wireless Tank Level Monitoring

IoT tank stage monitoring uses wireless sensors and devices to reveal the extent of liquid in a tank. This wi-fi structure reduces the complexity of the machine and makes studying processes more flexible. Horizontal scaling of the solution additionally will become smooth for industries and they gain the capability to behaviour immediately troubleshooting and take statistics-driven decisions. Moreover, hardware and implementation expenses are reduced, and groups advantage the capability to meet their changing enterprise requirements. A paper related to low-cost electric devices states the advantages of such mechanism and uses of a wireless based monitoring i.e., Wi-Fi along with the Arduino [1]. These documents present the concepts required for a good design implementing a wireless real-time framework feasible on a popular microcontroller called Arduino Uno. Another research paper sates the importance of a proper plan to avoid dry running of pumps. For this purpose, computing the level of water with the help of sensors fixed in the container or the place water is pumped from. In such a situation if such system senses the water level to be below required levels, then it aborts the pumping motor from functioning to ensure the well-being of the pumps and avoid dry running. An alerting sound or notification could be initiated in such cases alerting the user concerning the issue [2].

The use of GSM based technology have become popular today. Here, an advanced technique is brought forward which continuously keeps a record and is able to handle the functionalities of a system using a mobile network, in case of certain water level in containers like an overhead water tank, the operator can report to the registered module in the case of recognizing the particulars of a certain tank, also it is capable of regulating the pump automatically by turning OFF when the required amount of water has been filled up and sends the message back to the user, displaying the status of tank. Hence, its focused in managing with reinforcement of ultrasonic sensor accompanying the GSM technology [3].



Figure 2.1: Wireless Tank Level Monitoring

Here we have planned to build the IoT device controlled over the internet and there are prototype models for this topic, developed for detecting water levels by using the internet and various IoT devices. A central device like microprocessor connects to the internet and receives data and instructions to control sensors, with the help of a server capable of managing the users and their devices comprising of a compatible Android mobile application acting as a user interface [4]. Another author suggested a method assisting to realize the benefit of Bluetooth modules and how efficiently it can be produced as a compact device [5]

This system can measure liquid levels of not just tanks in fixed locations. It can be used to measure liquid levels in movable tanks as well (if no rapid movements are made causing large ripples). The liquid and water tank tracking system can also be implemented for tankers in transit. Through distant monitoring answers of IoT, a corporation can monitor water levels and quality of water in trucks carrying the fluid, along with the extent of fluid saved of their tankers in real-time. The water level monitoring device can be used to monitor liquid levels of not just water and since this system is in two separate components levels of other liquids in tanks can be measured also. IoT water tank stage tracking gadget may be used for any liquid like fuel, chemical compounds, and liquids; that too being conveyed in movable tankers having booths. (Laging, 2017)



Figure 2.2: Mobile Tanker Level Monitoring

A research paper indicated that one of the supervising systems which was generated for quantification of water levels, and it also uses ultrasonic sensing technology, PIC micro-controller, and GSM module as main devices. The ultrasonic sensor computes the free space from the sensor to the top of contained fluid. This system suggests the evolution of water measure monitoring system centred around the GSM module to notify the user via SMS (Short Message Service) when the liquid has reached minimum level and it will automatically switch OFF the electric water pump [6].

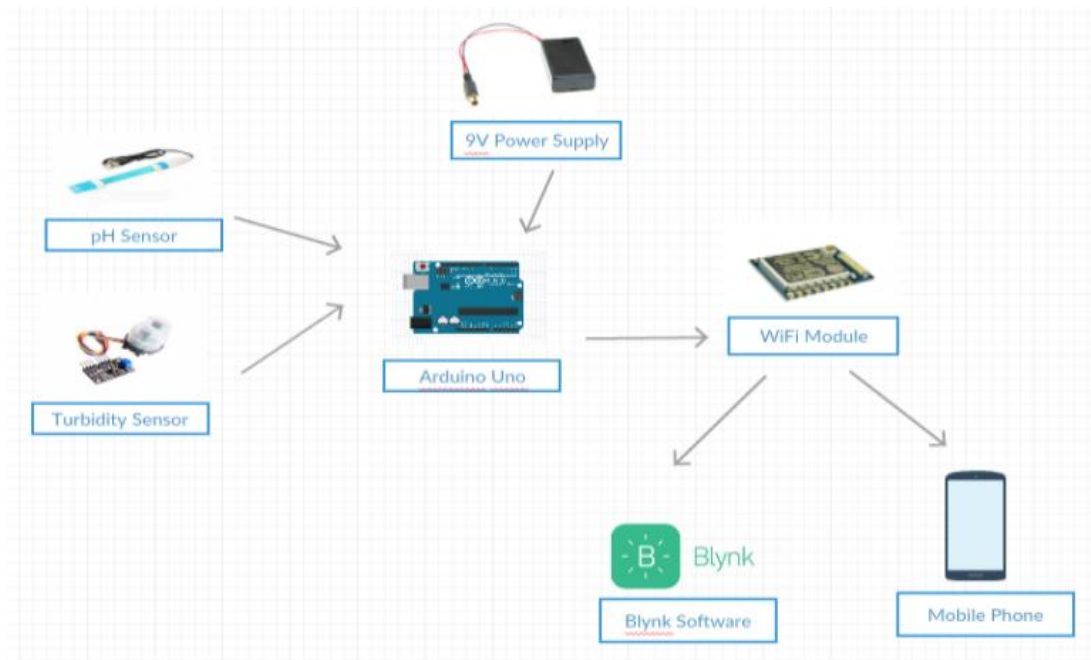


Figure 2.3: Water quality System Accessibility diagram

Figure above shows the topology of the IoT Water Quality Monitoring System in a diagram form. Arduino Uno will be used as the microcontroller for this project. The sensors needed for this project consist of a pH sensor and Turbidity sensor. The Wi-Fi module is used to send the results to the smartphone via Wi-Fi. The results were sent through Blynk software (may or may not be used), where the results will be shown on an android smartphone.

Chapter 3: Methodology

3.1 planning

Interviews and observations can be carried out to determine how many people are satisfied with the current water storage system that they have and what problems people face when storing of water. Also, the people can be interviewed on the proposed system and whether they will benefit from such a system and what useful feature that can be added to ease their work.

As the second stage of planning checking with books, websites, journal articles and research papers that allow you to get knowledge on new technology and innovations available at present, concerning the tank water level measuring system. This could be supportive to get a deep understanding on choosing of the appropriate technology for this venture.

Thirdly a minor planning stage is required to develop the web plication, here a study on suitable software and how features will be implemented are considered.

Finally, a proper testing and evaluation procedure must be planned to test the system for its accuracy and functionality.

3.2 Technologies to be used.

The project might be carried out using Arduino microcontrollers. Several sensors will be used to get statistics from the environment and water. The Arduino Uno board will be a suitable product to carry out this project and numerous sensors and modules will be used for gathering of all the records and data from the surroundings and water and convert them into usable information for the IoT device.

The proposed circuit diagrams are shown in figure below and these may be subjected to change depending on availability of products and compatibility at the time of making the final product.

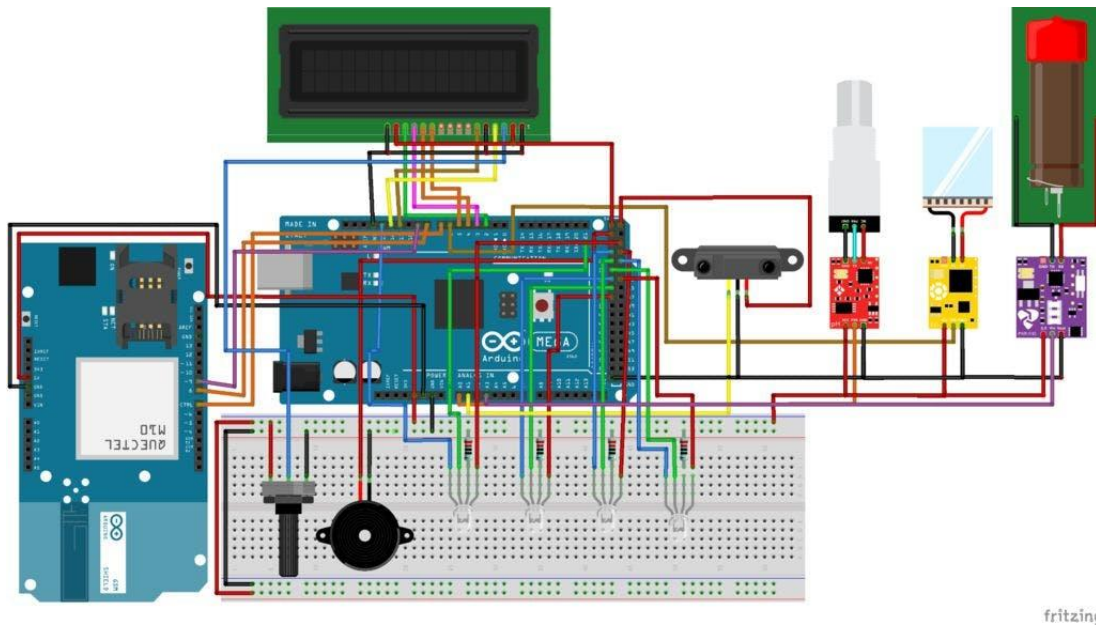


Figure 3.1: Circuit Diagram of Water Quality Measuring Component

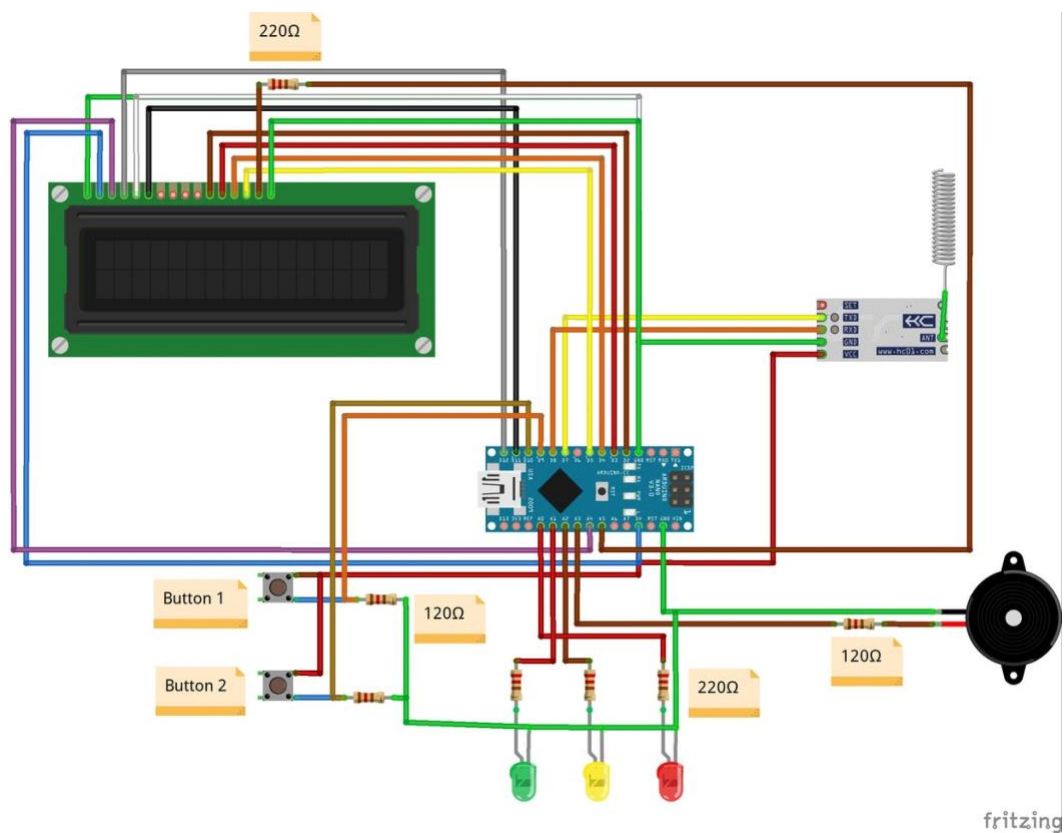


Figure 3.2: Circuit Diagram of Water Level Measuring Component

Certain devices that may be used for this project are mentioned below but it is not an exact list of the devices and may be subjected to change depending on compatibility and availability. The devices are listed as follows,

- NodeMCU

NodeMCU is an open-source IoT platform. It includes firmware that runs on the WIFI SOC from Espresso Systems and hardware which is primarily based at the ES module. The term "NodeMCU" via default refers to the firmware in place of the improvement kits. The firmware makes use of the Lue scripting language. It is based totally on the assignment and built at the Espresso Non-OS. It makes use of many open-source tasks, together with Launceston.

Advantage

The NodeMCU has a more powerful processor than the Arduino's Atmega328, a 4Mb flash memory, and lots more RAM. It essentially has almost the whole lot that an Uno has to offer, plus a great deal greater. It has fewer GPIO, ADC and PWM alternatives than the Uno, but supports Serial Communication protocols.



Figure 3.3: NodeMCU

- Sim 800 GSM IC CHIP, Sim800

SIM 800 GSM/GPRS SMT module is energy-packed in all its flexible features, packaged in a swish and compact design, supported with the Quadband the module gives you an international connectivity at your fingertip. SIM800 module gets along with embedded builders Industry too, its capabilities numerous embedded Interfaces presenting solutions to a utility like Remote Control/tracking, SMS manipulate, Data logging, etc., by way of the use of boost AT Commands. Besides low power consumption and rapid response, SIM 800 module gives get entry to many to manipulate capabilities like Bluetooth, USB, Serial interface and many others., imparting global, rapid time to market solutions to substantial Industry wishes in a count number of seconds. SIM800 is a quad-band GSM/GPRS module that works on frequencies GSM 850MHz, EGSM 900MHz and PCS 1900MHz. SIM800 capabilities GPRS multi-slot elegance 12/ elegance 10 (optional) and helps the GPRS coding schemes CS-1, CS-2, CS-3 and CS-four. A tiny configuration of SIM800 can meet nearly all the area necessities in users" programs, including M2M, clever phone, PDA and different cellular gadgets. SIM800 has sixty-eight SMT pads and presents all hardware interfaces among the module and customers" boards. (Wheeler, n.d.)



Figure 3.4: Sim 800 GSM IC CHIP, Sim800

Advantage

SIM 800 GSM Module. This GSM Modem can accept any GSM network operator SIM card and act similar to a cellular phone with its precise telephone number. The advantage of using this modem could be that you can use its RS232 port to talk and expand embedded packages.

- Ultrasonic sensor

All the ultrasonic transducer self-produced in KODO are highly stable. Maximized the adhesive strength by cutting a 0.5R groove on the circumference of the panel. Remote Control Function. Multiscreen Display. Alarm Function. Powerful Cavitation. Highlights: Providing Quick Response



Figure 3.5: Ultrasonic Sensor

Advantage

The ultrasonic sensor has an excessive frequency, high sensitivity, and high penetrating electricity therefore it could easily hit upon the outside or deep gadgets. These sensors have extra accuracy than other strategies for measuring the thickness and intensity of parallel surface

- Data transfer cable

A USB statistics transfer cable is useful to move data from an older laptop onto a more recent one. Newer operating structures are also packaged with software that understands the transfer cables and makes moving the documents smooth. Transfer cables can be used to transport documents from one laptop to every another.



Figure 3.6: Data Transfer Cable

Advantage

It is a solid replacement for old management systems and is the new manner facts may be maintained in a secure and at ease way. There is little possibility of mistakes being made, and

antique or obsolete information can be easily controlled and removed whilst statistics are transferred.

- Jumper Wires (Male to female wires)

Jumper wires usually are available in 3 versions: male-to-male, male-to-woman, and girl-to-girl. The distinction among every is in the long-run point of the cord. Male ends have a pin protruding and might plug into matters, while female ends do not longer and are used to plug things into. (Hemmings, 2018)

Advantage

Jumper wires typically come in 3 versions: male-to-male, male-to-girl, and woman-to-lady. The distinction among each is ultimately a factor of the cord. Male ends have a pin protruding and can plug into matters, while female ends do no longer and are used to plug things into.

Some of the other devices are as follows,

- Liquid Water PH Value Detection Sensor Control Board
- Bolt IoT Bolt Wi-Fi Module
- Arduino Mega Board
- RGB backlight positive LCD 16x2
- Turbidity Sensor
- Arduino UNO Board
- Arduino Nano with USB Cable
- Electric Motor
- Breadboard
- A buzzer
- Resistors (120 ohm, 220 ohm)
- LED Lights

3.3 User Requirements

Listed below are the requirements for users to successfully set up the system,

- Each house or company must have a suitable water tank
- Users must register to the system before using it.
- Users must have a suitable device to access the web application.
- Users must have suitable supply of water or electric power.

Listed below are the main requirements of the system,

- The system must indicate proper water levels and water quality measurements at real time
- The system must be assessable to users from any location at any time.

- The system must display pump status whether water is being pumped or not.
- System must provide alerting messages to users if water level is exceeding maximum capacity or is below required levels.
- The pumps should be able to switch on or off using the system.

The figure below shows how the system will be accessible to the user,

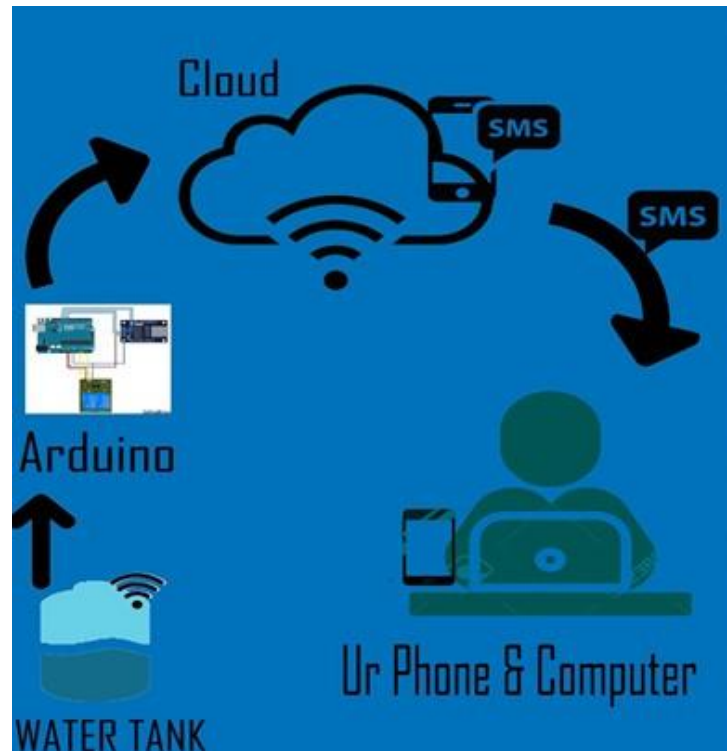


Figure 3.1 Accessibility Diagram

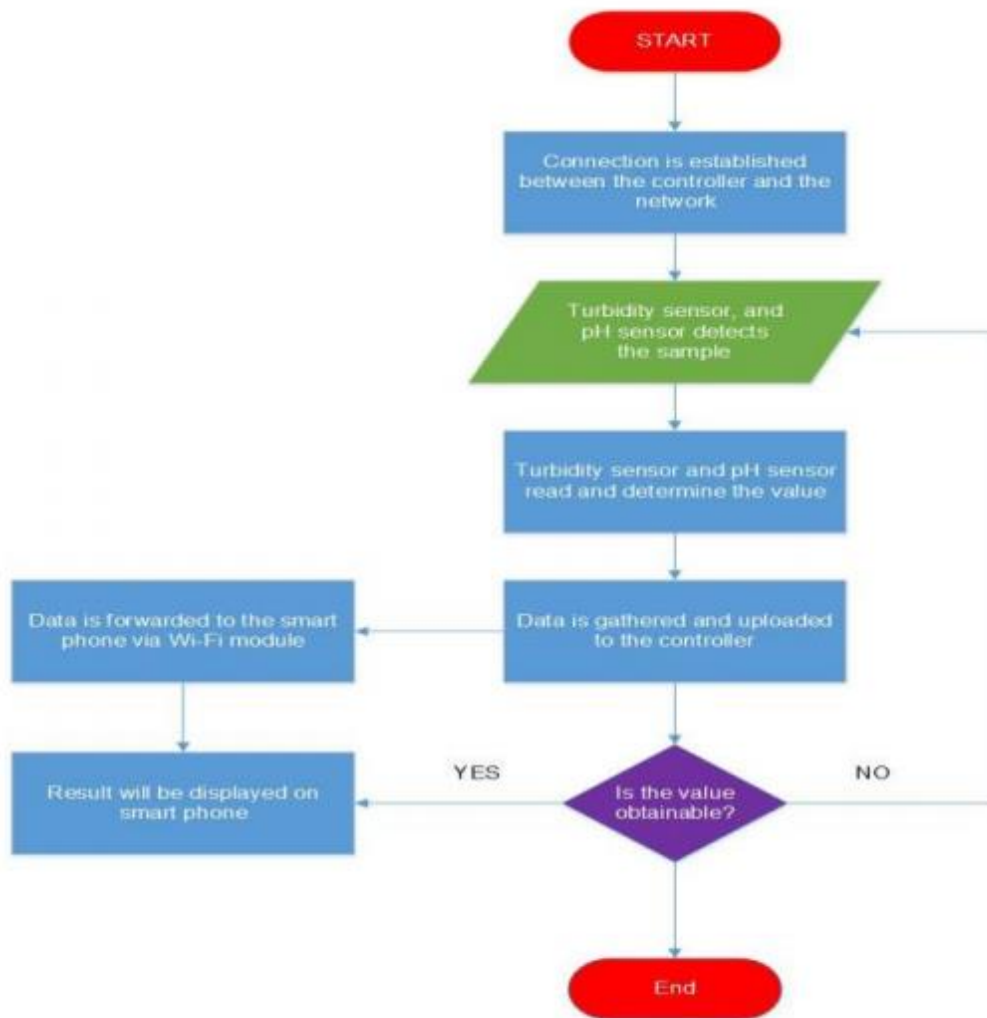


Figure 3.2: Flow chart water quality measuring component

As for the timeline of the project, a basic indication of the main stages of the project followed but the starting date and the ending date are shown in the following table. But these dates may be subjected to change due to availability of items and other such factors.

Timeline		
Objectives	Start date	End Date
Gathering the required equipment and sensors	15/07/2021	15/08/2021
Developing the Web application	15/08/2021	20/09/2021
Develop the hardware model for the water level components	20/9/2021	10/10/2021
Develop the hardware model for the water quality components	20/9/2021	10/10/2021
Connecting the web app with the hardware components	10/10/2021	12/10/2021
Testing the devices	12/10/2021	31/10/2021

Table 3.1: Timeline

3.4 Software Requirements

This project involves a number of software tools to be completed successfully. Software such as Google documents and Forms may be used to create forms and other such documents used for questionnaires and other tasks. Major software such as Visual Studio Code or Arduino IDE may be used for developing a platform and configuring of devices. Some of the software that may be used are mentioned below and may be subjected to change at the time of implementing the project.

- Blynk

Blynk is a hardware-agnostic IoT platform with white-label mobile apps, private clouds, device management, data analytics, and machine learning(may or may not be used).

- Arduino IDE 1.8.

The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. This software can be used with any Arduino board.

- Visual Studio Code

Visual Studio Code is a code editor redefined and optimized for building and debugging modern web and cloud applications (may or may not be used).

- Firebase/ Microsoft Azure

Firebase is a platform developed by Google for creating mobile and web applications, The Azure Internet of Things (IoT) is a collection of Microsoft-managed cloud services that connect, monitor, and control billions of IoT assets. These software might be used for this project depending on pricing or compatibility of the software with this project.

- Google Docs/Forms/Sheets

Google Documents, Forms or Sheets may be used for minor purposes of this project such as issuing questionnaires or writing and producing of written documents necessary for the assignment.

- Android studio

Android Studio may be used to build the web application required for this project but it will be considered depending on the other available software for this purpose.

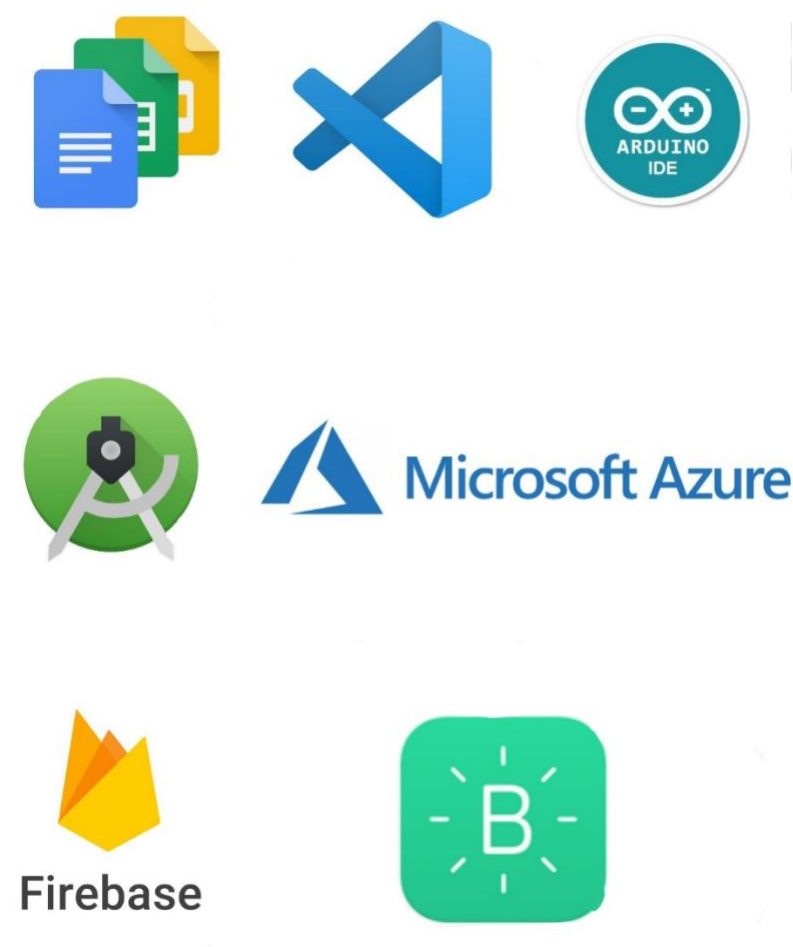


Figure 3.3: Software that may be used

Chapter 4: Testing and Evaluation

4.1 Testing

The testing process will be divided into two stages. The first stage of testing will test whether the components are functioning properly without any technical errors or whether the components are properly connected and compatible with each device. This stage includes testing of the Web Application and the code for both the web app and Arduino chips as well.

The second stage of testing will be used to determine the overall accuracy of the equipment and the compatibility of the system under real environmental factors.

In this stage the water level measuring component will be fixed to the top of the water tank and the water quality measuring component will be fixed to suitable location and the system must be registered and accessed. Once these steps are complete the testing process can begin.

Here we will use water from 3 water sources from three different locations and pump them into a tank which this system is set up.

Each sample of water will be pumped to the tank 5 times and tested at 5 random times. And the level of water and quality of water will be checked at these random intervals. The water quality measurement will be determined successful if all 5 tests give similar reading in the 5 attempts of testing for each of the 3 samples (It is notable that readings may slightly differ due to device imperfections).

The water level component will be determined successful if the indicated value in the system is similar to the actual value measured manually (manual water level can be calculated by measuring using meter ruler or other forms) during the 5 testing attempts for each of the 3 samples.

4.2 Evaluation

Upon successful completion of the testing process the system will be given to 3 domestic house owners who have tanks. Here 3 domestic house owners were selected because due to time limitations of this project there may not be sufficient time to evaluate using more people because the device has to be given to them one after another because only one device may be available.

Here each user will be using our system for 3 days and will evaluate the product on his personal ideas and thoughts. This includes his satisfaction towards overall use of the system, helpfulness and functionality, ease of use and if his requirements are met.

Chapter 5: Personnel and facilities

This project involves several people that have contributed to this project in certain ways. The names and their associated roles are mentioned under this section.

- Project Supervisor – Dr. Anuradha Jayakody, Senior Lecturer, Faculty of Computing, Computer Systems Engineering, Sri Lanka Institute of Information Technology.

- Project Team Members –
 - Name- Perera H. A.K.D
 - SLIIT ID- IT19360874
 - CURTIN ID- 20543912
 - Name- Kasthuriarachchi S.D.
 - SLIIT ID- IT19082066
 - CURTIN ID- 20543873

The tasks that will be undertaken by this project will be divided equally among the two members. The basic assignments of tasks are mentioned below and may change and vary depending on the problems and challenges faced during the project.

Task assignment is as follows,

Team Member 1 (Perera H.A.K.D)- Responsibilities and tasks include building of the proposed IoT system with the relevant Server configurations and design and building of the web application with the automatic notification or Alert system. This includes research on software that will be used and the compatibility with the software and hardware used as well. Minor tasks such as building of Google forms and handling of documents that may be needed for Questionnaires and other documents may be handled as well. Additional help towards building the water quality testing component may be provided as well.

Team Member 2 (Kasthuriarachchi S.D.) – Responsibilities and tasks include designing and building of the hardware component necessary for the proposed model. This includes research on proper devices used and test technologies to be selected. Hardware testing and compatibility checks are also one of the responsibilities. Minor tasks such as budget calculation and maintenance for hardware components may also be handled. Additional help towards the building of the web application may be provided as well.

Other personnel – This has several other which may help directly or indirectly to the project. This includes random people interviewed regarding the necessary features and feasibility of the proposed system and the 3 random people who would evaluate the system upon completion.

Chapter 6: Budget

The table below shows the budget allocated for this project. It should be noted that this is a rough budget evaluation and that it may be subjected to change. The budget is calculated with the current prices of devices used at the time of making of the proposal and miscellaneous expenses may also be added in the event of device failures, transportation and other activities that may have an effect on this project.

Hardware List

Total

LKR26,261.16

Item	Cost (LKR)
Liquid Water PH Value Detection Sensor Control Board	LKR3,800.00
Bolt IoT Bolt Wi-Fi Module	LKR3,323.77
Arduino Mega Board	LKR2,594.00
RGB backlight positive LCD 16x2	LKR2,531.93
Turbidity Sensor	LKR2,164.00
Arduino UNO Board	LKR2,120.00
SIM800C GSM GPRS Module	LKR1,390.00
Arduino Nano with USB Cable	LKR1,000.00
Ultrasonic Distance Sensor	LKR772.46
Electric Motor	LKR720.00
Breadboard	LKR335.00
Jumper Wires	LKR190.00
A buzzer	LKR155.00
Resistors (120 ohm, 220 ohm)	LKR120.00
LED (red)	LKR15.00
LED (yellow)	LKR15.00
LED (green)	LKR15.00
Miscellaneous Expenses	LKR5,000.00

Table 6.1: Budget

Chapter 7: References

- [1] J. Hartmanis and J. Van Leeuwen, Lecture Notes in Computer Science. .
- [2] G. Suciu, A. Vintea, S. C. Arseni, C. Butca, and V. Suciu, “Challenges and Solutions for Advanced Sensing of Water Infrastructures in Urban Environments,” pp. 349–352, 2015
- [3] B. Dhivyapriya, C. Gulabsha, S. P. Maniprabha, G. Kandasamy, and V. Chandrasekaran, “GSM BASED WATER TANK LEVEL MONITORING AND PUMP CONTROL,” pp. 108–112, 2016.
- [4] S. Jadhav, “Monitoring of Industrial Water Usage by using Internet of Things,” 2018 Int. Conf. Inf. , Commun. Eng. Technol., pp. 1–4, 2018.
- [5] J. Priya and S. Chekuri, “WATER LEVEL MONITORING SYSTEM USING IOT,” pp. 1813–1817, 2017.
- [6] S. M. K. Reza, S. A. Tariq, and S. M. M. Reza, “Microcontroller Based Automated Water Level Sensing and Controlling : Design and Implementation Issue,” vol. I, 2010